

BOOLEAN ALGEBRA IDENTITIES (LAWS)

1.	$A + A = A$	Idempotent law for or
2.	$A * A = A$	Idempotent law for and
3.	$A + B = B + A$	Commutative law for or
4.	$A * B = B * A$	Commutative law for and
5.	$A + (B + C) = (A + B) + C$	Associative law for or
6.	$A * (B * C) = (A * B) * C$	Associative law for and
7.	$A * (B + C) = (A * B) + (A * C)$	Distributive law for and over or
8.	$A + (B * C) = (A + B) * (A + C)$	Distributive law for or over and
9.	$A + 1 = 1$	Law of Union
10.	$A * 0 = 0$	Law of Intersection
11.	$A * (A + B) = A$	Law of Absorption
12.	$A + (A * B) = A$	Law of Absorption
13.	$A * 1 = A$	1 is the identity element for and
14.	$A + 0 = A$	0 is the identity element for or
15.	$\overline{\overline{A}} = A$	Double negative law
16.	$A + \overline{A} = 1$	Law of Complement for or
17.	$A * \overline{A} = 0$	Law of Complement for and
18.	$\overline{A + B} = \overline{A} * \overline{B}$	DeMorgan's Law
19.	$\overline{AB} = \overline{A} + \overline{B}$	DeMorgan's Law
20.	$A \oplus B = A \overline{B} + \overline{A} B$	Exclusive or (xor)
21.	$\overline{A \oplus B} = \overline{A} \oplus B = A \oplus \overline{B}$	Not xor
22.	$A + \overline{A} B = A + B$	Extra no name law
23.	$(A + B)(A + C) = A + BC$	Extra no name law